

STEMI

Date

December 17, 2025 (Corrected XPath and SAS code)

May 27, 2025 (updated to ignore eSituation.10 - Other Associated Symptoms if it has a Pertinent Negative, "Symptom Not Present")

August 26, 2021

Definition

The patient experienced a STEMI.

Criteria Description

Patient care reports where:

- any symptom or impression is STEMI
or
- cardiac rhythm is STEMI
or
 - any symptom or impression is angina pectoris or unspecified angina
and
 - cardiac rhythm is left bundle branch block
- or
- STEMI protocol is used
or
- a STEMI destination team pre-arrival alert or activation is performed.

Pseudocode

Patient care reports where:

eSituation.09 - Primary Symptom

eSituation.10 - Other Associated Symptoms (without Pertinent Negative)

eSituation.11 - Provider's Primary Impression

eSituation.12 - Provider's Secondary Impressions

- I21.0* ST elevation (STEMI) myocardial infarction of anterior wall
- I21.1* ST elevation (STEMI) myocardial infarction of inferior wall
- I21.2* ST elevation (STEMI) myocardial infarction of other sites

- I21.3 ST elevation (STEMI) myocardial infarction of unspecified site

OR

eVitals.03 - Cardiac Rhythm

- STEMI Anterior Ischemia
- STEMI Inferior Ischemia
- STEMI Lateral Ischemia
- STEMI Posterior Ischemia

OR

(

eSituation.09 - Primary Symptom

eSituation.10 - Other Associated Symptoms (without Pertinent Negative)

eSituation.11 - Provider's Primary Impression

eSituation.12 - Provider's Secondary Impressions

- I20 Angina pectoris
- I20.0 Unstable angina
- I20.9 Angina pectoris, unspecified

AND

eVitals.03 - Cardiac Rhythm

- Left Bundle Branch Block

)

OR

eProtocols.01 Protocols Used

- Medical-ST-Elevation Myocardial Infarction (STEMI)

OR

eDisposition.24 - Destination Team Pre-Arrival Alert or Activation

- Yes-STEMI

XPath Code

/EMSDataset/Header/PatientCareReport

```
[
  eSituation/(eSituation.09, eSituation.10[not(@PN)], eSituation.11, eSituation.12)
  [matches(., '^((I21.0)|(I21.1)|(I21.2)|(I21.3))')]
or
  eVitals/eVitals.VitalGroup/eVitals.03 =
```

```

    ('9901051', '9901053', '9901055', '9901057')
or
(
    eSituation/(eSituation.09, eSituation.10[not(@PN)], eSituation.11,
        eSituation.12) = ('I20', 'I20.0', 'I20.9')
    and
    eVitals/eVitals.VitalGroup/eVitals.03 = '9901021'
)
or
eProtocols/eProtocols./ProtocolGroup/eProtocols.01 = '9914143'
or
eDisposition/eDisposition.HospitalTeamActivationGroup/eDisposition.24 = '4224013'
]

```

SAS Code

The NEMSIS public release research dataset does not include the Pertinent Negative attribute on eSituation.10 - Other Associated Symptoms, which was added in NEMSIS version 3.5, so the SAS code will select some records where a symptom was actually documented as “Symptom Not Present.” This is estimated to affect less than 3% of v3.5 records.

```

proc sql;
    create table StemiKey as
    select distinct pcrKey from
    (
        select pcrKey from nemsis.factPcrPrimarySymptom where
            prxmatch('/^((I21.0)|(I21.1)|(I21.2)|(I21.3))/', eSituation_09)
        union
        select pcrKey from nemsis.factPcrAdditionalSymptom where
            prxmatch('/^((I21.0)|(I21.1)|(I21.2)|(I21.3))/', eSituation_10)
        union
        select pcrKey from nemsis.factPcrPrimaryImpression where
            prxmatch('/^((I21.0)|(I21.1)|(I21.2)|(I21.3))/', eSituation_11)
        union
        select pcrKey from nemsis.factPcrSecondaryImpression where
            prxmatch('/^((I21.0)|(I21.1)|(I21.2)|(I21.3))/', eSituation_12)
        union
        select pcrKey from nemsis.factPcrVital where pcrVitalKey in
            (
                select vitalKey from nemsis.pcrVitalEcgGroup where eVitals_03 in
                    ('9901051', '9901053', '9901055', '9901057')
            )
        union
        select pcrKey from nemsis.factPcrVital where pcrVitalKey in

```

```

(
  select vitalKey from nemsis.pcrVitalEcgGroup where eVitals_03 =
    '9901021'
)
and pcrKey in
(
  select pcrKey from nemsis.factPcrPrimarySymptom where
    eSituation_09 in ('I21', 'I20.0', 'I20.9')
  union
  select pcrKey from nemsis.factPcrAdditionalSymptom where
    eSituation_10 in ('I21', 'I20.0', 'I20.9')
  union
  select pcrKey from nemsis.factPcrPrimaryImpression where
    eSituation_11 in ('I21', 'I20.0', 'I20.9')
  union
  select pcrKey from nemsis.factPcrSecondaryImpression where
    eSituation_12 in ('I21', 'I20.0', 'I20.9')
)
union
select pcrKey from nemsis.factPcrProtocol where eProtocol_01 = '9914143'
union
select pcrKey from nemsis.factPcrDestinationTeam where eDisposition_24 =
  '4224013')
;
quit;

```

Discussion

None.

References

NEMSIS V3 Public STEMI Dashboard. Global Inclusion Criteria and Filters.
wiki.utahdcc.org/confluence/display/NTD/V3+Public+STEMI+Dashboard.